PRODUCT REQUIREMENTS DOCUMENT

INDmoney SIP Discovery Engine

A public-evidence system for explaining why first-time Indian mutual-fund investors lapse by month 12 — plus a retention-solution appendix.

Version 1.0 · Draft for review · October 2026
Owner: Growth / Product Discovery
Evidence snapshot: 8,115 documents · 1,726 strict candidates · 1,726 deep-coded · 334 grounded signals
Public sources only. No internal analytics, CRM records, or primary interviews.

Document control

Field	Value
Document	INDmoney SIP Discovery Engine — Product Requirements Document
Version	1.0
Status	Draft for review
Owner	Growth / Product Discovery
Reviewers	Growth, Product, Data/Analytics, Compliance, Customer Support
Primary metric	At least one successful instalment in the last 30 days, 12 months after the first SIP
Evidence base	8,115 public documents across six sources; 1,726 deep-coded; 334 grounded signals
Constraints	Public data only; no fund/stock recommendations; no fear, FOMO, or urgency-based nudges
Related artefacts	AI Evidence Log (PDF/CSV/JSONL), Streamlit dashboard, case deck (PPTX/PDF), case write-up (PDF)

Change log

Version	Date	Change
0.1	Day 1	Scope, sources, data model, and codebook v0.1.
0.2	Day 2	Collectors, strict filter, extraction pipeline, grounding rules.
0.3	Day 3	Prompt v0.8, segment evidence, theme taxonomy, opportunity scoring.
0.4	Day 4	Dashboard, evidence log, validation harness, red-team tests.
1.0	Day 5	Case deck, case write-up, PRD, retention appendix, final QA.

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1. Executive summary

INDmoney can see who lapses. It cannot yet see the moment that made the investor stop. This PRD specifies a discovery engine that reads public investor conversations, extracts structured signals about SIP behaviour, grounds every code in a verified quote, and quantifies which moments are most associated with a first-year lapse. The output is not another sentiment dashboard; it is an evidence base for choosing retention plays.

The engine answers nine discovery questions, from who first-time investors are to what would make them stay. It combines LLM extraction with deterministic Python/SQL for counts, prevalence, statistics, and opportunity scoring. Every behavioural quote must be an exact substring of the source document; unsupported codes are dropped by a grounding layer that checks first-person actions, negations, attempts, payment-mechanism stops, temporary pauses, and segment evidence.

What the evidence says

- The strongest lapse-associated theme is cash-flow shocks and competing commitments: 14 signals, 100% lapse rate among resolved outcomes.
- System friction is the most fixable: failed mandates/debits and cancellation mazes create lapses the investor did not choose.
- Market noise is high-reach but not uniformly harmful; many investors continue through drawdowns when they have a goal and a habit.
- The protective theme is equally important: goals, streaks, community, and human support keep SIPs alive.
- The evidence is public, self-selected, and correlational. It is a hypothesis-generation and prioritisation tool, not a causal estimate.

What to build

- Pilot 1 — Mandate rescue: pre-debit health check, three-day repair window, auto-retry, and a plain-language reason.
- Pilot 2 — Pause, don't stop: a 1–3 month pause with auto-resume, step-down amount, and salary-cycle alignment.
- Pilot 3 — A cancellation path that works: pause beside cancel, a status tracker, and a named support owner.
- Instrument the 12-month metric and the pause-to-resume bridge before scaling any play.

2. Problem and opportunity

2.1 The problem

A first-time SIP is a habit built on a promise: a small amount, a fixed date, and a long horizon. The habit breaks in month 3, 7, or 11 for reasons that are rarely captured in a transaction log. The investor may have lost a job, missed a salary cycle, lost trust after a failed mandate, tried to cancel and failed, or simply concluded that the fund was not worth it. The product sees the missing instalment; it does not see the story behind it.

Public feedback is an underused source of that story. App reviews capture friction at the moment it happens; Reddit and forums capture reasoning, comparisons, and peer advice; YouTube comments capture reactions to market narratives. Used carefully, these sources can tell us which moments to instrument and which interventions to test. They cannot tell us the true lapse rate or the size of a segment.

2.2 The opportunity

The opportunity is not a better fund recommendation. It is keeping the SIP alive through the moments that break the habit: a failed debit, a cash-flow shock, a market drawdown, a confusing cancellation flow. These are workflow, clarity, and trust problems. They are solvable without giving investment advice, without fear-based nudges, and without waiting for a market recovery.

3. Goals, non-goals, and success metrics

3.1 Goals

- Answer all nine discovery questions with quote-level evidence.
- Collect 5,000–8,000 public documents; screen to 1,500–2,500 strict SIP/mutual-fund candidates; deep-code as many as the free-tier quota allows, with transparent reporting of the achieved number (currently 1,726).
- Build a 200-item stratified gold set, run a second model, compute Cohen's kappa, and review disagreements.
- Run a red-team suite covering third-person false positives, negation, hypotheticals, payment-mechanism stops, temporary pauses, sarcasm, Hinglish, quote truncation, source bias, and prompt injection.
- Ship a working dashboard, an AI Evidence Log (PDF/CSV/JSONL), a case write-up, a deck (PDF/PPTX), and this PRD.

3.2 Non-goals

- Causal estimation of lapse drivers. The engine generates hypotheses; experiments establish causality.
- Individual-level prediction or scoring of named users.
- Fund, stock, or product recommendations. The engine may compare mechanics and categories, never recommend a specific fund.
- Replacing internal analytics, CRM, or primary interviews. Public evidence complements them.
- Using fear, FOMO, urgency, or dark patterns to improve retention.

3.3 Engine success metrics

Metric	Definition	Target
Coverage	Share of strict candidates deep-coded and reported	>= 60% of strict candidates, with the remainder screened
Quote fidelity	Share of behavioural quotes that pass exact/normalized verification	>= 95%
Grounding precision	Share of kept codes supported by the quote after deterministic checks	>= 90% on the audited sample
Inter-model agreement	Cohen's kappa on theme presence and outcome	>= 0.60; report confidence interval

Metric	Definition	Target
Time to answer	Time from a question to a filtered evidence view	< 10 minutes in the dashboard
Reproducibility	A clean run reproduces counts and exports from the same corpus snapshot	Deterministic counts; model calls logged with prompt version

4. Users and jobs-to-be-done

User	Job-to-be-done	What they need
Growth PM	Choose the next retention sprint	Ranked opportunities, evidence quotes, effort/impact
Product manager	Design a specific intervention	Moments, triggers, current workarounds, guardrails
Research lead	Defend the method	Sampling, codebook, kappa, red-team results, limitations
Data analyst	Reproduce and extend	Schema, queries, exports, run logs
Compliance / support	Check the claims and the copy	Quote provenance, no advice, no fear/FOMO, escalation paths

5. Discovery questions and evidence

The nine questions are the contract between the engine and the Growth team. Each answer must point to a theme, a signal count, and a quote. The table below is the current read; the dashboard is the live view.

#	Question	Current evidence-backed answer	Confidence
1	Who are first-time investors?	Salaried, often 20s–30s, starting small; segment cues are sparse and self-reported.	Low–medium
2	What expectations do they bring?	Steady/FD-like or market-beating returns; the first flat period is an expectation shock.	Medium
3	What happens in the first 30 days?	KYC, mandate, first debit, and NAV/date confusion; confusion and anxiety are common coded emotions.	Medium
4	What triggers pause/reduce/stop?	Cash-flow shocks, job changes, competing goals, drawdowns, app/cancellation friction.	Medium–high
5	How do volatility and news affect continuation?	They raise checking and anxiety for some; others continue through drawdowns with a goal and a habit.	Medium
6	How do cash-flow shocks and competing goals matter?	Cash-flow shocks are a strong lapse-associated theme: the SIP was affordable last month, not this month. Competing goals add a second claim on the same money.	Medium–high (small n)
7	How do app, mandate, debit, cancellation, and support failures affect trust?	They create system-forced lapses and erode control; the investor often cannot tell what happened.	High (friction), low (outcome)
8	Which segments are most at risk?	Directionally, early-stage and cash-flow-constrained investors; app reviews overrepresent friction and segment cues are sparse.	Low
9	What makes them continue or restart, and what should INDmoney prioritise?	Pause instead of stop, mandate rescue, goal progress, human support, expectation-setting.	Medium

6. Scope, sources, and sampling

6.1 In scope

- Public posts and reviews from Play Store, App Store, Reddit, YouTube, ValuePickr, and TradingQ&A.;
- First-person accounts of starting, paying, missing, pausing, stopping, restarting, increasing, reducing, or reviewing a SIP or mutual-fund investment in India.
- Expectations, emotions, help-seeking, app/mandate/debit/cancellation/support friction, segment cues, and self-reported outcomes.

- English and Hinglish text; Devanagari is detected as Hindi, but Hinglish may be labelled English.

6.2 Out of scope

- Stocks, IPOs, F&O;, crypto, US stocks, and trading-only experiences.
- Generic app praise/complaints with no personal SIP or mutual-fund experience.
- Fund recommendations, third-party advice, news, explainers, promotions, and 'you should invest' posts.
- Internal analytics, CRM records, support tickets, or primary interviews.

6.3 Source mix and bias

Source kind	Documents	Signals	Known outcomes	Lapse rate	Bias note
app_store	2,617	81	9	89%	Short, friction-heavy, rating-driven.
forum	199	6	0	n/a	Long-form, expert/peer audience, selection bias toward active investors.
play_store	4,877	122	10	50%	Short, friction-heavy, often written right after a problem.
reddit	340	122	62	81%	Reflective, peer-influenced, overrepresents engaged investors.
youtube	82	3	0	n/a	Reactive, market-narrative-driven, low context.

Date range: 2014-09-19 to 2026-10-19. 8,115 documents collected; 3,565 keyword-relevant; 1,726 strict SIP/mutual-fund candidates; 1,726 deep-coded; 334 grounded signals across 332 documents.

Deep-coding is still in progress for Play Store and part of App Store. The source table shows signals available in the current snapshot, not a final yield. Play Store has the largest document count but the lowest extraction coverage at this snapshot.

7. System architecture

The engine has two layers. The probabilistic layer reads unstructured text and proposes structured codes. The deterministic layer verifies quotes, grounds codes, computes counts, assigns themes, scores opportunities, and exports evidence. The model never computes prevalence, statistics, or the final ranking.

Layer	Component	Responsibility	Output
Collect	Source collectors	Play Store, App Store, Reddit, YouTube, ValuePickr, TradingQ&A	Raw documents
Normalize	Text cleaner	HTML entities, whitespace, URLs, deduplication, language flag	Normalized text
Screen	Strict filter	SIP/mutual-fund relevance, personal-experience cues, exclusion rules	Strict candidates
Extract	LLM adapter	Batched JSON extraction with Pydantic schema, retries, checkpointing, token budget	Proposed signals
Verify	Quote verifier	Exact/normalized substring match; reject fabricated or over-long quotes	Verified quotes
Ground	Deterministic rules	First-person, negation, attempt, payment-stop, temporary-pause, segment checks	Grounded signals
Analyze	Python/SQL	Theme assignment, prevalence, lapse rate, lift, opportunity score	Tables and exports
Serve	Streamlit + exports	Dashboard, CSV/JSONL/PDF evidence log	Decision artefacts

Storage: SQLite for development, with a Postgres-compatible schema for deployment on Neon/Vercel. The current DB is data/indmoney.db; the schema is defined in src/models.py.

8. Data model

Entity	Key fields	Purpose
Source	id, name, kind, url, meta	Provenance and source-level filters

Entity	Key fields	Purpose
Document	id, source_id, external_id, title, raw_text, normalized_text, published_at, language, meta	One public post/review/comment
Signal	id, document_id, journey_stage, triggers, behaviors, emotions, expectations, help_seeking, outcome, segment, quote, confidence, raw	One grounded evidence unit
Run	id, kind, model, prompt_version, status, stats, started_at, finished_at	Reproducibility and token accounting
GoldItem	id, document_id, strata, labels, reviewer, status	200-item validation set
Theme / Opportunity	derived at analysis time	Ranking and export; not hand-edited

Document.meta stores the extract_run_id and extract_prompt_version, so analysis can select the latest signal per document and resume extraction without reprocessing. Signal.raw preserves the segment quote, grounding warnings, and reground status for audit.

9. Pipeline and quality controls

9.1 Collection

Each collector writes raw and normalized text, source metadata, external ID, published date, language, and a content hash for deduplication. Reddit uses old.reddit JSON with an RSS fallback; ValuePickr and TradingQ&A; use Discourse /search.json and /t/{id}.json; YouTube uses the public page and comment continuation where available. Collectors are run sequentially to avoid SQLite write locks.

9.2 Strict screening

The strict filter requires a SIP/mutual-fund term and a personal-experience cue, then removes trading, stock, IPO, crypto, US-stock, generic app-bug, and third-party advice. The filter is intentionally high-recall; the LLM relevance gate makes the final call and is itself evaluated.

9.3 Extraction

The extractor sends batches of documents with the compact codebook. It requests strict JSON, validates with Pydantic, verifies quotes, grounds codes, and writes signals. Batch failures are logged and do not stop the run. A token budget guard stops the run before the free-tier limit; checkpoints allow --resume. Model routing is provider-agnostic through LiteLLM, with Gemini 3.5 Flash-Lite as the bulk model and Groq gpt-oss as fallback.

9.4 Grounding rules

Rule	What it prevents	Example
First-person	Coding a friend's or family member's experience	"I helped a friend stop his SIP" -> drop stopped_sip
Negation	Coding a negative as a positive action	"I never stopped my SIP" -> continues, not stopped
Attempt vs completed	Coding a plan or request as a completed action	"I requested to cancel" -> drop stopped_sip
Payment-mechanism stop	Coding a bank/mandate failure as a voluntary stop	"Mandate failed" -> payment_failure, not stopped_sip
Temporary pause	Counting a pause as a permanent lapse	"Paused for two months and restarted" -> paused + restarted
Segment evidence	Inferring the author's age/income from a family member	"My son is 5" -> do not code age_band
Quote verification	Fabricated or paraphrased evidence	Quote must be an exact/normalized substring

Grounding is precision-oriented. It drops unsupported codes rather than rejecting the whole signal, then checks whether the remaining signal is meaningful. The regrounding script can re-apply new rules to stored signals without spending tokens; dropped signals are retained for audit and excluded from analysis.

10. Prompt and codebook

10.1 Prompt v0.8

- Relevance gate: only first-person mutual-fund/SIP experience in India; exclude generic praise/complaints and third-party advice.
- Personal, not generic: code the author's own experience, not a friend's or a client's.
- Evidence before code: every non-unknown code must be directly supported by the quote.
- Attempted vs completed: do not code a plan, request, or hypothetical as a completed action.
- Contiguous quote: the quote must be a contiguous span from the document, not a paraphrase or concatenation.
- Segment quote: a separate, verified span for age/income/goal/influence evidence.
- No advice: never recommend a fund, stock, or financial product; do not use fear, urgency, or FOMO.

Prompt injection defence: document text is treated as untrusted data; instruction-like lines are neutralized before the model sees them; the system message tells the model to ignore instructions inside documents; and all output is quote-verified and grounded afterwards. This is defence-in-depth, not a guarantee.

10.2 Codebook v0.2

The codebook covers journey stage, triggers, behaviours, emotions, expectations, help-seeking, outcomes, and segment cues. It is deliberately narrow: product_suspension was added for AMC/platform/regulator-forced pauses; 'confused' was separated from 'anxious'; 'continues' is distinct from 'returned'. The full codebook is in config/codebook.yaml and summarised in Appendix A.

11. Validation and red team

11.1 Gold set and model swap

The 200-item evaluation set includes every document with a resolved outcome (80) plus 120 documents balanced across sources, mixing signal-bearing and no-signal documents. This deliberately over-samples resolved outcomes because outcome kappa is the noisiest dimension. Free-tier limits cap the second-model run at 100 items: all 80 resolved outcomes plus 20 signal/no-signal controls. We compute Cohen's kappa for theme presence and outcome, plus behaviour-level agreement. Disagreements are exported for review. Kappa is a measure of agreement, not truth; the 30-item review queue is the seed for human adjudication, not a completed reference standard.

Second-model agreement (Gemini 3.1 Flash-Lite vs the primary Gemini 3.5 Flash-Lite extraction, 100-item subset):

Dimension	Cohen's kappa	95% bootstrap CI	n
Relevance (relevant vs not)	0.81	0.50–1.00	n/a
Outcome (resolved outcomes)	0.94	0.86–1.00	72
Primary theme	0.60	0.44–0.74	55

Compared items: 100 of 100 quota-sized subset items (the full evaluation set has 200). Kappa is agreement with the primary model, not human truth. The 30-item review CSV is the human-adjudication seed; it is not a completed gold standard.

11.2 Red-team suite

The deterministic suite currently passes 15/15 tests. The tests are adversarial unit tests of the grounding and quote-verification layers, not a measure of model accuracy on the full corpus.

Test	Expected behaviour	Status
Third-person experience	Do not code the friend's stop as the author's	Pass
Hypothetical action	Do not code 'if I stop' as stopped	Pass
Negation	'I never stopped' -> continues, not stopped	Pass
Payment-mechanism stop	Code payment_failure, not voluntary stopped_sip; keep outcome stopped	Pass
Temporary pause	Normalize to paused/restarted, not stopped	Pass

Test	Expected behaviour	Status
Attempted cancellation	Do not code a request as completed	Pass
Segment family member	Do not infer the author's age from a child's age	Pass
Valid completed stop	Retain the stop when first-person and explicit	Pass
Quote exact match	Accept exact substring	Pass
Quote whitespace match	Accept normalized, flag as soft	Pass
Fabricated quote	Reject	Pass
Prompt injection	Sanitize instruction-like lines	Pass
Normal text unchanged	Do not over-sanitize	Pass
Empty payload	Drop signals with no codes	Pass

Known limitations: public posts are self-selected; self-reported outcomes are not the 12-month metric; associations are not causal; app reviews are short; Hinglish/Indic coverage is limited; segment cues are sparse; quote verification prevents fabrication, not misinterpretation; model-swap agreement is not human ground truth; the corpus spans 2014-2026. Free-tier quotas forced the second model to Gemini 3.1 Flash-Lite: an initial Gemini 3.8 Flash run stopped at 36 items when its 20-request daily cap was reached. The reported 100-item swap uses one model throughout, not a mix.

12. Dashboard and evidence log

The Streamlit dashboard has five views: Overview, Theme deep dive, Signal explorer, Segments and stages, and Method and guardrails. Filters include source, theme, outcome, journey stage, confidence, and search. Every chart is computed from the grounded signal table; the signal explorer shows the exact quote, source, date, codes, and grounding warnings. A CSV download supports offline review.

The AI Evidence Log exports CSV, JSONL, and a PDF. The PDF includes the method, theme summaries, and a sample of verified quotes with source provenance. The log is designed for a reviewer who wants to check the evidence without opening the database.

12.1 Evidence-log fields

Field	Description
signal_id / document_id	Stable IDs for audit
source / source_kind	Platform and source name
published_at / language	Time and language context
quote	Verified contiguous span from the document
journey_stage / triggers / behaviors	Grounded codes
emotions / expectations / help_seeking	Grounded codes
outcome / segment	Self-reported outcome and segment cues
themes	Deterministic theme assignment
confidence / warnings	Model confidence and grounding warnings

13. Opportunity scoring

The opportunity score is a prioritisation heuristic, not a causal effect size. It combines reach (share of signal-bearing documents), severity (lapse lift vs the corpus baseline), evidence strength (signal count and confidence), solvability (how much of the moment INDmoney can influence), and effort (implementation drag).

Formula: score = 100 × (0.35 × reach + 0.30 × severity + 0.20 × evidence + 0.15 × solvability) × (1 - 0.25 × effort). For themes with no resolved outcome, severity uses a modest friction prior (0.35) rather than zero, and the theme is flagged n/a; the deck and dashboard never present the baseline placeholder as an observed lapse rate.

The score is sensitive to the corpus mix. App reviews overrepresent friction; Reddit/forums overrepresent engaged investors. Use the score to choose what to investigate, then validate with internal data and experiments.

14. Retention-solution appendix

This appendix turns each theme into a testable retention play. It is deliberately solution-agnostic about funds: the interventions change the workflow, clarity, and support around the SIP, not the investment product. Every play must pass the guardrails in Section 16: no fear, no FOMO, no urgency, no specific fund recommendation, no dark patterns.

14.1 Prioritisation

Priority	Play	Theme	Why now	Primary metric
1	Mandate rescue	Failed debits / mandate breakage	High fixability; a lapse the investor did not choose	Mandate recovery rate
2	Pause, don't stop	Cash-flow shock; cancellation friction	Preserves the habit through a temporary squeeze	Pause-to-resume within 90 days
3	Cancellation path that works	Can't stop or fix it	Restores control and reduces complaint-driven churn	Cancellation save rate
4	Goal-progress framing	Market noise; expectation gap	Changes the meaning of a drawdown without fear	12-month retention
5	First-statement explainer	Expectation gap; product confusion	Sets expectations before the first flat period	30-day statement comprehension

14.2 Failed debits and mandate breakage

Evidence snapshot: 59 signals across 59 documents; lapse rate n/a; lift n/a; opportunity score 70.7. unresolved outcome; friction prior.

Problem. The debit failed, not the investor.

The moment. The investor sees a missed instalment, a bank message, or no message at all. They do not know whether the SIP is still active, whether the money will be debited again, or what to fix.

Interventions.

- Pre-debit health check 3 days before the debit date: mandate status, bank account status, balance signal, and a one-tap repair.
- Auto-retry within the same cycle with a plain-language reason and a fallback payment method.
- A status card: scheduled, debited, failed, retrying, repaired, or paused; one owner and one response-time promise.

Success metric. Mandate recovery rate; failed-debit-to-next-successful-instalment; support contacts per 1,000 failed debits.

Experiment. Holdout 10%; randomize eligible failed mandates to the rescue flow vs standard retry; 8-week read; guardrail on complaints and duplicate debits.

Guardrails. No specific fund or stock recommendation; no fear, FOMO, or urgency; clear opt-out; no dark patterns; support escalation available; monitor complaints and unintended pauses.

14.3 Can't stop or fix it

Evidence snapshot: 38 signals across 38 documents; lapse rate 89%; lift 1.15x; opportunity score 65.8. empirical lapse lift.

Problem. I tried to stop it, but it didn't stop.

The moment. The investor wants control. The flow is slow, unclear, or fails; money is debited again; support does not resolve it. Trust drops faster than the SIP.

Interventions.

- Pause beside cancel, with a clear effective date and one-tap resume.
- Cancellation status tracker: requested, effective, refunded, closed; a named owner and a response-time promise.
- A 'what happens next' explainer: remaining instalments, mandate, and refund timeline.

Success metric. Cancellation save rate; time-to-effective; repeat contacts; complaint rate.

Experiment. A/B the pause-beside-cancel flow vs current cancel; 6-week read; guardrail on unwanted pauses and opt-out clarity.

Guardrails. No specific fund or stock recommendation; no fear, FOMO, or urgency; clear opt-out; no dark patterns; support escalation available; monitor complaints and unintended pauses.

14.4 What keeps SIPs alive

Evidence snapshot: 49 signals across 49 documents; lapse rate 0%; lift n/a (protective); opportunity score 53.8. empirical lapse lift.

Problem. What kept you in?

The moment. Investors who continue describe a goal, a habit, a support system, or a reason to stay. These are protective factors, not just absence of friction.

Interventions.

- Celebrate goal progress and streak milestones without FOMO or urgency.
- A 'you stayed invested through…' recap that reinforces the identity, not the return.
- Community stories and human encouragement at the moments that usually trigger a stop.

Success metric. 12-month retention; streak/recap engagement; referral and community participation.

Experiment. Randomize the recap vs standard; 8-week read; guardrail on gambling-like streak mechanics.

Guardrails. No specific fund or stock recommendation; no fear, FOMO, or urgency; clear opt-out; no dark patterns; support escalation available; monitor complaints and unintended pauses.

14.5 Cash-flow shocks and competing commitments

Evidence snapshot: 14 signals across 14 documents; lapse rate 100%; lift 1.29x; opportunity score 52.5. empirical lapse lift.

Problem. The SIP was affordable last month, not this month.

The moment. A salary delay, job change, medical bill, or family expense makes the next debit feel unsafe. The investor's only visible option is 'stop'.

Interventions.

- Pause 1–3 months with auto-resume, no penalty, and a clear resume date.
- Step-down amount (for example, 50% for 3 months) that preserves the habit and the goal link.
- Salary-cycle alignment and a job-loss grace window; an emergency-fund nudge that is educational, not a product pitch.

Success metric. Pause-to-resume within 90 days; step-down uptake; 12-month retention; support load.

Experiment. Randomize eligible at-risk investors to pause/step-down vs cancel-only; 12-week read; segment by salary date and tenure.

Guardrails. No specific fund or stock recommendation; no fear, FOMO, or urgency; clear opt-out; no dark patterns; support escalation available; monitor complaints and unintended pauses.

14.6 Which fund, which goal?

Evidence snapshot: 6 signals across 6 documents; lapse rate 100%; lift 1.29x; opportunity score 43.1. empirical lapse lift.

Problem. Which fund, which goal?

The moment. The investor has a house, wedding, or education goal and does not know whether the SIP fits it. The result is a stop, not a reallocation.

Interventions.

- One goal per SIP, with a goal-progress view and a plain-language fit check.

- House, wedding, education, and retirement planners that show the trade-off between time horizon and volatility.
- A pause-and-resume path for short-term goals instead of stopping the habit.

Success metric. Goal-tagged SIP share; 12-month retention; goal-review completion; support contacts.

Experiment. Randomize goal-tagging + review vs standard onboarding; 12-week read; segment by goal type.

Guardrails. No specific fund or stock recommendation; no fear, FOMO, or urgency; clear opt-out; no dark patterns; support escalation available; monitor complaints and unintended pauses.

14.7 Market noise and loss aversion

Evidence snapshot: 31 signals across 31 documents; lapse rate 55%; lift 0.70x; opportunity score 42.9. empirical lapse lift.

Problem. The portfolio went red and the app went silent.

The moment. A drawdown feels like a verdict on the decision. Without a goal frame, the investor checks daily, reads noise, and stops to relieve anxiety.

Interventions.

- Drawdown explainer that shows goal progress, time horizon, and what the current value means in context.
- A 'what happens if you stop' calculator framed as a trade-off, not a threat.
- An opt-in human call or community story for investors who are panicking; no fear-based push notifications.

Success metric. 12-month retention; daily-check frequency; panic-related support contacts; pause vs stop mix.

Experiment. Randomize new investors to goal-progress framing vs return-only framing; 6-month read; pre-register guardrails on anxiety and complaints.

Guardrails. No specific fund or stock recommendation; no fear, FOMO, or urgency; clear opt-out; no dark patterns; support escalation available; monitor complaints and unintended pauses.

14.8 Returns didn't match the promise

Evidence snapshot: 3 signals across 3 documents; lapse rate n/a; lift n/a; opportunity score 24.0. unresolved outcome; friction prior.

Problem. Returns didn't match the promise.

The moment. The investor expected steady growth; the first flat or negative period feels like a broken product. The gap is set at onboarding and reinforced by the first statement.

Interventions.

- Expectation-setting at onboarding: a range of outcomes, time horizon, and risk, in plain language.
- First-statement explainer separating XIRR, absolute return, and market noise; a 'why your portfolio is down' checklist.
- A next-step prompt that is educational, not a product pitch.

Success metric. 30-day statement comprehension; 12-month retention; complaint rate; support contacts after the first statement.

Experiment. Randomize the explainer vs current statement; 6-week read; guardrail on confusion and complaint rate.

Guardrails. No specific fund or stock recommendation; no fear, FOMO, or urgency; clear opt-out; no dark patterns; support escalation available; monitor complaints and unintended pauses.

14.9 I don't know what I bought

Evidence snapshot: 1 signals across 1 documents; lapse rate n/a; lift n/a; opportunity score 21.7. unresolved outcome; friction prior.

Problem. I don't know what I bought.

The moment. Direct vs regular, NAV, fund category, and expense ratio are invisible or confusing. The investor cannot explain the product to themselves.

Interventions.

- A 'what am I invested in' statement with direct/regular and expense-ratio clarity.
- A plain-language comparison of mechanics and categories, never a specific fund recommendation.
- A human callback or glossary for first-time investors.

Success metric. Comprehension score; 30-day support contacts; 12-month retention; opt-out rate.

Experiment. A/B the statement + glossary vs current; 6-week read; guardrail on information overload.

Guardrails. No specific fund or stock recommendation; no fear, FOMO, or urgency; clear opt-out; no dark patterns; support escalation available; monitor complaints and unintended pauses.

14.10 Paused by the system, not the investor

Evidence snapshot: 7 signals across 7 documents; lapse rate 50%; lift 0.64x; opportunity score 19.7. empirical lapse lift.

Problem. The system paused me, not the other way around.

The moment. An AMC, platform, or regulator restriction stops the SIP. The investor is not told why, for how long, or what happens to the instalment.

Interventions.

- Proactive notice with the reason, expected duration, and the exact impact on the instalment and mandate.
- A clear alternative route or category option, without recommending a specific fund.
- Auto-resume when the restriction lifts, with confirmation and no penalty.

Success metric. Share of forced pauses with a same-day notice; auto-resume rate; complaint rate; 12-month retention.

Experiment. Staged rollout by restriction type; compare notice + auto-resume vs notice-only; 8-week read.

Guardrails. No specific fund or stock recommendation; no fear, FOMO, or urgency; clear opt-out; no dark patterns; support escalation available; monitor complaints and unintended pauses.

14.11 Out of sight, out of mind

Evidence snapshot: 5 signals across 5 documents; lapse rate 50%; lift 0.64x; opportunity score 15.1. empirical lapse lift.

Problem. Out of sight, out of mind.

The moment. No immediate problem; the SIP simply fades. The investor misses a debit, then another, and the habit is gone.

Interventions.

- A monthly goal-progress recap, not a market-noise feed.
- A two-missed-debit re-engagement sequence with a one-tap repair.
- A low-effort 'SIP health score' that makes the next action obvious.

Success metric. Two-missed-debit recovery; 12-month retention; re-engagement conversion.

Experiment. Randomize the recap + health score vs standard; 8-week read; guardrail on notification fatigue.

Guardrails. No specific fund or stock recommendation; no fear, FOMO, or urgency; clear opt-out; no dark patterns; support escalation available; monitor complaints and unintended pauses.

15. Experiment and measurement plan

15.1 Metric hierarchy

Level	Metric	Definition	Cadence
Primary	12-month retention	At least one successful instalment in the last 30 days, 12 months after the first SIP	Monthly cohort
Secondary	3-month activation	Three successful instalments in the first 90 days	Monthly
Secondary	6-month continuation	At least one successful instalment in month 6	Monthly
Secondary	Pause-to-resume	Share of paused SIPs that resume within 90 days	Monthly
Secondary	Mandate recovery	Share of failed/at-risk mandates repaired before the next debit	Weekly
Secondary	Cancellation save	Share of cancellation intents that choose pause, step-down, or repair	Weekly
Guardrail	Complaints	Complaints per 1,000 active SIPs; themes: unauthorized debit, mis-selling, dark pattern	Weekly
Guardrail	Opt-outs	Share of users who disable notifications or close the SIP after a nudge	Weekly
Guardrail	Support load	Contacts per 1,000 active SIPs; median resolution time	Weekly

15.2 Experiment design

- Unit: eligible investor-SIP pairs, randomized at the moment of risk (failed debit, cancellation intent, pause intent, or first drawdown).
- Design: two-arm A/B with a 10% holdout; pre-register the primary metric, guardrails, and minimum detectable effect.
- Duration: 8–12 weeks for leading indicators; 12 months for the primary retention metric. Do not declare success on a leading indicator alone.
- Power: size the experiment for the smallest segment that matters; if underpowered, use a staggered rollout with a matched comparison.
- Analysis: intention-to-treat; report confidence intervals; segment by tenure, source, salary date, and risk cohort; pre-register subgroups.
- Guardrails: stop if complaints, unauthorized debits, or opt-outs exceed the threshold; review copy with compliance before launch.

Internal data is required for measurement. The discovery engine uses public data only; the experiment layer uses internal event data under the company's privacy and access controls.

16. Risks, ethics, and guardrails

Risk	Impact	Mitigation
Self-selected public data	Overstates friction and underrepresents silent lapses	Triangulate sources; label as directional; validate with internal data
Source bias	App reviews overrepresent friction; Reddit overrepresents engaged investors	Report by source; do not average without weights; use narrative sources for explanation
Model error	False positives/negatives in codes	Quote verification, grounding, gold set, kappa, red team, human adjudication
Prompt injection	Model follows instructions inside a document	Sanitize instruction-like lines; treat as data; quote verification; monitor
Privacy	Public posts may contain personal details	Do not reproduce handles; redact in exports; retain only public content; honor deletion requests
Regulatory / advice	Copy could look like investment advice	No fund/stock recommendations; compliance review; educational framing
Dark patterns	Pause/cancel flows could manipulate	Pause beside cancel; clear effective dates; no hidden fees; opt-out parity
Causal overreach	Prioritising a theme that does not cause lapse	Call it association; experiment before scaling; pre-register
Time span	2014–2026 platform and market changes confound themes	Report date distribution; segment by year; avoid long-run comparisons

17. Timeline and milestones

Day	Milestone	Output
1	Scope, sources, schema, collectors	8,115 documents; SQLite DB; source config
2	Strict filter, extraction pipeline, codebook v0.2	1,726 strict candidates; prompt v0.8; grounding
3	Deep extraction, theme taxonomy, opportunity scoring	1,726 deep-coded; 334 signals
4	Dashboard, evidence log, gold set, model swap, red team	Working dashboard; PDF/CSV/JSONL evidence log; validation report
5	Case write-up, deck, PRD, final QA	PDF/PPTX deck; case write-up; PRD with retention appendix

Current status (honest)

- Deep-coded: 1,726 documents, covering every strict SIP/MF candidate and falling inside the 1,500–2,500 target.
- Signals: 334 across 332 documents. Narrative sources (Reddit, forums, YouTube) carry more explanatory weight than app reviews.
- Dashboard, evidence log, model swap, and red team are built; the 30-item human-adjudication queue is the remaining validation step before treating kappa as ground truth.
- No paid API credits were used; keys are stored in a gitignored .env and are never printed.

18. Acceptance criteria and reproduction

18.1 Acceptance criteria

- The engine answers all nine discovery questions with a theme, a signal count, and at least one verified quote.
- Every behavioural quote is an exact/normalized substring of the source; fabricated quotes are rejected.
- The dashboard loads without errors and supports source, theme, outcome, stage, confidence, and text filters.
- The evidence log exports CSV, JSONL, and PDF with provenance and warnings.
- The gold set, kappa, model swap, and red-team results are reported, including failures and limitations.
- The deck and PRD include the 12-month metric, the retention appendix, and the no-fear/no-FOMO/no-advice guardrails.
- The reproduction steps work from a clean environment with the same corpus snapshot and API keys in .env.

18.2 Reproduction

- Create a virtual environment and install requirements: python -m venv .venv; .venv\Scripts\pip install -r requirements.txt.
- Copy .env.example to .env and add GEMINI_API_KEY and GROQ_API_KEY. Never commit .env.
- Collect: python scripts/collect_sample.py (or the source-specific collectors).
- Extract: python scripts/extract_signals.py --source reddit --limit 400 --batch-size 6 --resume.
- Analyze: python scripts/analyze.py; export: python scripts/export_evidence_log.py.
- Dashboard: streamlit run dashboard/app.py.
- Validate: python scripts/build_gold_set.py; python scripts/model_swap.py; python scripts/compare_model_swap.py; python scripts/red_team.py.
- Deliverables: python scripts/build_charts.py; python scripts/build_deck.py; python scripts/build_case_writeup.py; python scripts/build_prd.py.

Appendix A. Codebook summary

The full codebook is in config/codebook.yaml. The summary below is for reviewers; the model receives a compact version.

Dimension	Codes
Journey stage	pre_start, m0_setup, m1_first_debit, m2_3_early, m4_6_mid, m7_12_late, beyond_12, unknown
Triggers	market_drawdown, fund_underperformance, job_change, income_shock, expense_shock, life_event, goal_competition, liquidity_need, trust_break, app_friction, mandate_issue, bank_issue, payment_failure, product_suspension, advice_or_peer, tax_or_regulation, other
Behaviours	started_sip, increased_sip, decreased_sip, paused_sip, stopped_sip, restarted_sip, switched_product, withdrew, sought_help, checked_frequently, other
Emotions	anxious, confused, regret, distrust, reassured, confident, frustrated, indifferent, other
Expectations	high_returns, regular_income, capital_safety, liquidity, goal_fit, transparency, low_fees, other
Help-seeking	searched_online, asked_forum, contacted_support, asked_friend, used_advisor, no_help, other
Outcomes	continues, increased, reduced, paused, stopped, withdrew, returned, unknown
Segments	age_band, income_band, goal_type, influence_source, city_tier, occupation, unknown

Appendix B. Source list

Source	Kind	Access	What it is good for	Bias
Play Store	App reviews	Public review pages	Friction at the moment; volume	Short, problem-heavy
App Store	App reviews	Public RSS/pages	Friction at the moment; volume	Short, rating-driven
Reddit	Forum/community	old.reddit JSON; RSS fallback	Reasoning, comparisons, peer advice	Engaged, self-selected
YouTube	Video comments	Public pages	Reactions to market narratives	Low context, reactive
ValuePickr	Investor forum	Discourse /search.json	Long-form investor reasoning	Expert/active investors
TradingQ&A	Investor forum	Discourse /search.json	Long-form investor reasoning	Expert/active investors

Appendix C. Evidence-log schema

Column	Type	Example
signal_id	integer	128
document_id	integer	4021
source	string	reddit
source_kind	string	reddit
published_at	ISO date	2025-11-02
language	string	en
quote	text	I stopped my SIP after 3 months because I lost my job.
journey_stage	string	m2_3_early
triggers	list	job_change, income_shock
behaviors	list	stopped_sip
outcome	string	stopped
segment	json	{"age_band": "25-29"}
themes	list	cashflow_shock
confidence	string	high
warnings	list	trigger job_change derived from quote